

# Evaluation Metrics

# Introduction

- The most important task in building any machine learning model is to evaluate its performance
  - Evaluation metrics
  - Using different metrics for performance evaluation, we should be able to improve our model's overall predictive power before we roll it out for production on unseen data.
  - Without doing a proper evaluation of the Machine Learning model by using different evaluation metrics, and only depending on accuracy, can lead to a problem when the respective model is deployed on unseen data and may end in poor predictions.
- Machine learning Tasks: Supervised Learning and Un- supervised Learning
  - Ex: Classification and Regression

- ## Confusion Matrix

- Confusion Matrix is a performance measurement for the machine learning classification problems where the output can be two or more classes. It is a table with combinations of predicted and actual values.
- ***A confusion matrix is defined as the table that is often used to describe the performance of a classification model on a set of the test data for which the true values are known.***
- It is extremely useful for measuring the Recall, Precision, Accuracy, and AUC-ROC curves.

|                 | Predicted Positive  | Predicted Negative  |
|-----------------|---------------------|---------------------|
| Actual Positive | True Positive (TP)  | False Negative (FN) |
| Actual Negative | False Positive (FP) | True Negative (TN)  |

# Evaluation Metrics

- **Accuracy**

- Accuracy measures how often the classifier correctly predicts. We can define accuracy as the **ratio of the number of correct predictions and the total number of predictions.**

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

- Use case: Out of all the patients who visited the doctor, how many were correctly diagnosed as Covid positive and Covid negative.
- Accuracy is useful when the target class is **well balanced** but is not a good choice for the unbalanced classes.
- **Imagine the scenario where we had 99 images of the dog and only 1 image of a cat present in our training data.**
- **Then our model would always predict the dog, and therefore we got 99% accuracy.**
- In reality, Data is always imbalanced; for example Spam email, credit card fraud, and medical diagnosis.
- Hence, if we want to do a better model evaluation and have a full picture of the model evaluation, other metrics such as recall and precision should also be considered.

- **Precision**

- Precision explains how many of the correctly predicted cases actually turned out to be positive.
- In other words, precision tells out of all that were marked as positive, how many are actually truly positive.

$$Precision = \frac{TruePositive}{TruePositive + FalsePositive}$$

- Precision is useful in the cases where False Positive is a higher concern than False Negatives.
- Use case: “Out of all the people predicted as COVID-positive, how many actually had COVID?”

Suppose a hospital uses a model to detect COVID-19 from patient data.

- The model predicts 100 people as COVID-positive
- Out of those, 80 actually have COVID
- The remaining 20 are false alarms (false positives)

$$Precision = \frac{80}{80 + 20} = 0.8 = 80\%$$

**Precision matters when false positives are costly or problematic.**

**In airport screening:**

- You don't want to wrongly flag too many healthy travelers
- High precision → fewer unnecessary quarantines

- **Recall (Sensitivity or True Positive Rate)**

- Recall explains how many of the actual positive cases we were able to predict correctly with the model.

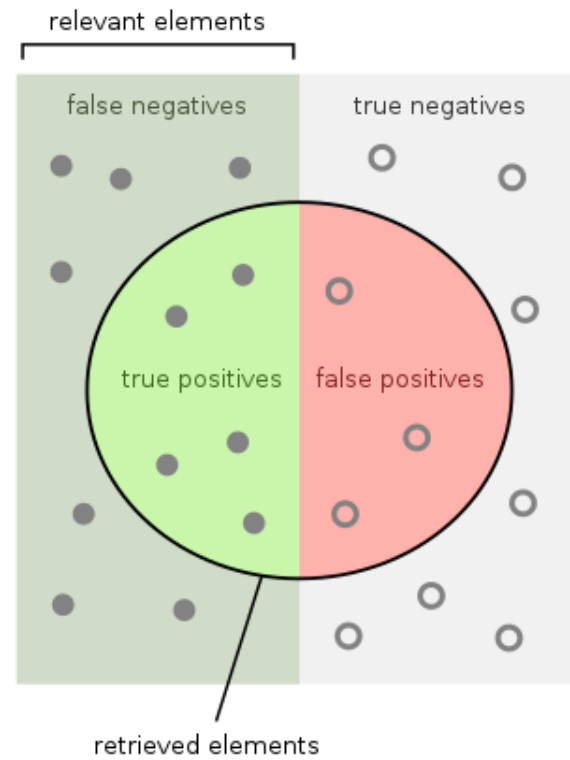
$$\text{Recall} = \frac{\text{TruePositive}}{\text{TruePositive} + \text{FalseNegative}}$$

- It is a useful metric in cases where False Negative is of higher concern than False Positive.
- *I.e.*, Recall matters when **missing a positive case is dangerous**.

- A false negative means:

- An infected person is told they are healthy (ex: rabies)
- They continue normal activities
- They may die without timely medical intervention
- So, high recall ensures most infected people are identified.

- *It is important in medical cases where it doesn't matter whether we raise a false alarm but the actual positive cases should not go undetected!*



How many retrieved items are relevant?

$$\text{Precision} = \frac{\text{true positives}}{\text{true positives} + \text{false positives}}$$

How many relevant items are retrieved?

$$\text{Recall} = \frac{\text{true positives}}{\text{true positives} + \text{false negatives}}$$

# F1 Score

- It gives a combined idea about Precision and Recall metrics. It is maximum when Precision is equal to Recall.
- **F1 Score is the harmonic mean of precision and recall.**

$$F1 = 2. \frac{Precision \times Recall}{Precision + Recall}$$

- F1 Score could be an effective evaluation metric in the following cases:
  - When FP and FN are equally costly.
  - Adding more data doesn't effectively change the outcome
  - True Negative is high



| n=165          | Predicted:<br>NO | Predicted:<br>YES |     |
|----------------|------------------|-------------------|-----|
| Actual:<br>NO  | TN = 50          | FP = 10           | 60  |
| Actual:<br>YES | FN = 5           | TP = 100          | 105 |
|                | 55               | 110               |     |

- **Accuracy:** Overall, how often is the classifier correct?
  - $(TP+TN)/total = (100+50)/165 = 0.91$
- **Recall or True Positive Rate:** When it's actually yes, how often does it predict yes?
  - $TP/actual\ yes = 100/105 = 0.95$
  - also known as "Sensitivity" or "Recall"
- **Precision:** When it predicts yes, how often is it correct?
  - $TP/predicted\ yes = 100/110 = 0.91$
- **True Negative Rate:** When it's actually no, how often does it predict no?
  - $TN/actual\ no = 50/60 = 0.83$
  - equivalent to  $(1 - \text{False Positive Rate})$
  - also known as "Specificity"

| n=165 |  | Predicted:<br>NO | Predicted:<br>YES |          |
|-------|--|------------------|-------------------|----------|
|       |  | Actual:<br>NO    | TN = 50           | FP = 10  |
|       |  |                  |                   | 60       |
|       |  | Actual:<br>YES   | FN = 5            | TP = 100 |
|       |  |                  |                   | 105      |
|       |  |                  | 55                | 110      |

# Q

- A hospital develops a machine learning model to detect COVID-19. The model is tested on 200 patients. Out of these, 80 patients actually have COVID-19 and 120 do not. The model correctly identifies 70 COVID-positive patients, incorrectly labels 20 healthy patients as COVID-positive
- Compute the **accuracy, precision, recall, F1 score** of the model.

# Q

- A hospital develops a machine learning model to detect COVID-19. The model is tested on 200 patients. Out of these, 80 patients actually have COVID-19 and 120 do not. The model correctly identifies 70 COVID-positive patients, incorrectly labels 20 healthy patients as COVID-positive
- Compute the **accuracy, precision, and recall** of the model.
- Given:
  - $TP = 70$
  - $FP = 20$
  - $FN = 10$
  - $TN = 100$
  - $Total = 200$

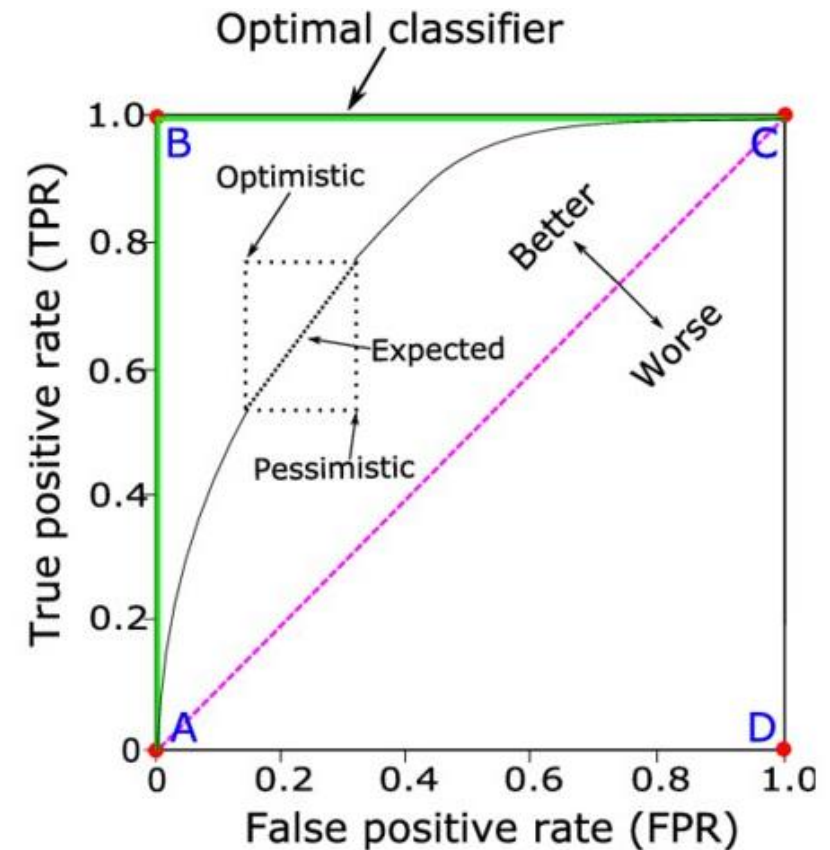
## Q2

- A spam detection system classifies emails as **Spam** or **Not Spam**. During evaluation on a test dataset:
  - Total number of emails = 500
  - Number of actual spam emails = 200
  - The model correctly identified 160 spam emails
  - The model incorrectly classified some non-spam emails as spam = 40
- Compute the **accuracy, precision, and recall** of the model.

- Total emails = 500
- Actual Spam (Positive) = 200 → Actual Not Spam = 300
- True Positives (TP) = 160
- False Positives (FP) = 40
- False Negatives (FN) = 200 – 160 = 40
- True Negatives (TN) = 300 – 40 = 260

- **AUC-ROC**

- The Receiver Operator Characteristic (ROC) is a probability curve that plots the TPR(True Positive Rate) against the FPR(False Positive Rate) at various threshold values and separates the 'signal' from the 'noise'.
- The **Area Under the Curve (AUC)** is the measure of the ability of a classifier to distinguish between classes. From the graph, we simply say the area of the curve ABDE and the X and Y-axis.
- From the graph shown below, the greater the AUC, the better is the performance of the model at different threshold points between positive and negative classes.
  - This simply means that When AUC is equal to 1, the classifier is able to perfectly distinguish between all Positive and Negative class points.
  - When AUC is equal to 0, the classifier would be predicting all Negatives as Positives and vice versa.
  - When AUC is 0.5, the classifier is not able to distinguish between the Positive and Negative classes.



# Appendix



- Ex: Disease diagnosis
  - **Sensitivity** (recall) refers to a test's ability to designate an individual with disease as positive.
  - A highly sensitive test means that there are few false negative results, and thus fewer cases of disease are missed.
  - The **specificity** of a test is its ability to designate an individual who does not have a disease as negative.

# Confusion Matrix for Multiclass Classification

The confusion matrix for the IRIS dataset

Example: Setosa

| Actual Values | Predicted Values |                |                |
|---------------|------------------|----------------|----------------|
|               | Setosa           | Versicolor     | Virginica      |
| Setosa        | 16<br>(cell 1)   | 0<br>(cell 2)  | 0<br>(cell 3)  |
| Versicolor    | 0<br>(cell 4)    | 17<br>(cell 5) | 1<br>(cell 6)  |
| Virginica     | 0<br>(cell 7)    | 0<br>(cell 8)  | 11<br>(cell 9) |

**TP:** The actual value and predicted value should be the same. So concerning Setosa class, the value of cell 1 is the TP value.

**FN:** The sum of values of corresponding rows except the TP value

$$FN = (\text{cell 2} + \text{cell 3}) = (0 + 0) = 0$$

**FP :** The sum of values of corresponding column except the TP value.

$$FP = (\text{cell 4} + \text{cell 7}) = (0 + 0) = 0$$

**TN:** The sum of values of all columns and row except the values of that class that we are calculating the values for.

$$TN = (\text{cell 5} + \text{cell 6} + \text{cell 8} + \text{cell 9}) = 17 + 1 + 0 + 11 = 29$$

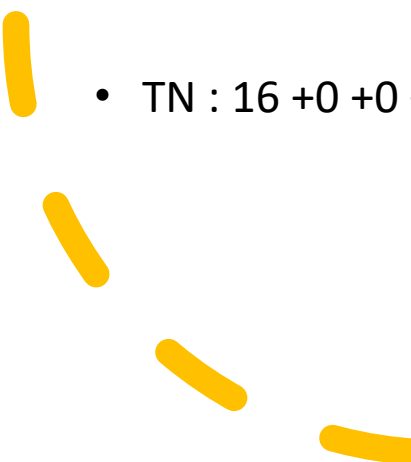


- Similarly, for **Versicolor** class the values/ metrics are calculated as below:

- TP : 17 (cell 5)

- FN :  $0 + 1 = 1$  (cell 4 +cell 6)

- FP :  $0 + 0 = 0$  (cell 2 + cell 8)



- TN :  $16 + 0 + 0 + 11 = 27$  (cell 1 + cell 3 + cell 7 + cell 9).

# Confusion Matrix for Multiclass Classification

|                 |        | True Class |        |       |
|-----------------|--------|------------|--------|-------|
|                 |        | Apple      | Orange | Mango |
| Predicted Class | Apple  | 7          | 8      | 9     |
|                 | Orange | 1          | 2      | 3     |
|                 | Mango  | 3          | 2      | 1     |

For example, if we take class Apple, then let's see what are the values of the metrics from the confusion matrix.

- **$TP = 7$**

- **$TN = (2+3+2+1) = 8$**

- **$FP = (8+9) = 17$**

- **$FN = (1+3) = 4$**

- **$Precision = 7/(7+8+9) = 0.29$**

- **$Recall = 7/(7+4) = 0.64$**

- **$F1-score = 0.40$**



|        |          |           |        |      |      |
|--------|----------|-----------|--------|------|------|
| Actual | Elephant | 25        | 3      | 0    | 2    |
|        | Monkey   | 3         | 53     | 2    | 3    |
|        | Fish     | 2         | 1      | 24   | 2    |
|        | Lion     | 1         | 0      | 2    | 71   |
|        |          | Elephant  | Monkey | Fish | Lion |
|        |          | Predicted |        |      |      |

- Compute the Average Precision and Average Recall of the Classifier

|                                                                                |          | CONDITION<br>determined by "Gold Standard"                                                                 |                                                                                                    |                                                                                     |                                                                                |
|--------------------------------------------------------------------------------|----------|------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| TOTAL POPULATION                                                               |          | CONDITION POS                                                                                              | CONDITION NEG                                                                                      | PREVALENCE<br>$\frac{\text{CONDITION POS}}{\text{TOTAL POPULATION}}$                |                                                                                |
| TEST<br>OUT-<br>COME                                                           | TEST POS | True Pos<br>TP                                                                                             | Type I Error<br>False Pos<br>FP                                                                    | Precision<br>Pos Predictive Value<br>$\text{PPV} = \frac{\text{TP}}{\text{TEST P}}$ | False Discovery Rate<br>$\text{FDR} = \frac{\text{FP}}{\text{TEST P}}$         |
|                                                                                | TEST NEG | Type II Error<br>False Neg<br>FN                                                                           | True Neg<br>TN                                                                                     | False Omission Rate<br>$\text{FOR} = \frac{\text{FN}}{\text{TEST N}}$               | Neg Predictive Value<br>$\text{NPV} = \frac{\text{TN}}{\text{TEST N}}$         |
| ACCURACY<br>ACC<br>$\text{ACC} = \frac{\text{TP} + \text{TN}}{\text{TOT POP}}$ |          | Sensitivity (SN), Recall<br>Total Pos Rate<br>TPR<br>$\text{TPR} = \frac{\text{TP}}{\text{CONDITION POS}}$ | Fall-Out<br>False Pos Rate<br>FPR<br>$\text{FPR} = \frac{\text{FP}}{\text{CONDITION NEG}}$         | Pos Likelihood Ratio<br>LR +<br>$\text{LR} + = \frac{\text{TPR}}{\text{FPR}}$       | Diagnostic Odds Ratio<br>DOR<br>$\text{DOR} = \frac{\text{LR} +}{\text{LR} -}$ |
|                                                                                |          | Miss Rate<br>False Neg Rate<br>FNR<br>$\text{FNR} = \frac{\text{FN}}{\text{CONDITION POS}}$                | Specificity (SPC)<br>True Neg Rate<br>TNR<br>$\text{TNR} = \frac{\text{TN}}{\text{CONDITION NEG}}$ | Neg Likelihood Ratio<br>LR -<br>$\text{LR} - = \frac{\text{TNR}}{\text{FNR}}$       |                                                                                |

# References

- [Frontiers | Sensitivity, Specificity, and Predictive Values: Foundations, Pliabilities, and Pitfalls in Research and Practice \(frontiersin.org\)](#)

[Confusion Matrix for Your Multi-Class Machine Learning Model](#)  
[| by Joydwip Mohajon | Towards Data Science](#)

|              |          | Predicted Class                            |                                                            |                                                                                 |
|--------------|----------|--------------------------------------------|------------------------------------------------------------|---------------------------------------------------------------------------------|
|              |          | Positive                                   | Negative                                                   |                                                                                 |
| Actual Class | Positive | True Positive (TP)                         | False Negative (FN)<br><b>Type II Error</b>                | <b>Sensitivity</b><br>$\frac{TP}{(TP + FN)}$<br>Or Recall or True Positive Rate |
|              | Negative | False Positive (FP)<br><b>Type I Error</b> | True Negative (TN)                                         | <b>Specificity</b><br>$\frac{TN}{(TN + FP)}$<br>True Negative Rate              |
|              |          | <b>Precision</b><br>$\frac{TP}{(TP + FP)}$ | <b>Negative Predictive Value</b><br>$\frac{TN}{(TN + FN)}$ | <b>Accuracy</b><br>$\frac{TP + TN}{(TP + TN + FP + FN)}$                        |